

MISHIMA, Japan

WMO: 476570

Lat: 35.1139N

Lon: 138.9253E

Elev: 22

StdP: 101.06

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.1	-0.9	-11.9	1.4	6.2	-8.9	1.8	5.1	8.8	9.3	8.1	9.7	1.1	120	0.370

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.3	33.1	25.5	32.1	25.1	31.1	24.7	26.4	31.1	25.9	30.3	25.4	29.6	3.8	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.2	20.3	29.1	24.6	19.7	28.7	24.1	19.0	28.3	82.6	31.1	80.3	30.4	78.1	29.7	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.7	6.5	5.5	-4.5	35.1	1.1	0.8	-5.2	35.7	-5.9	36.1	-6.5	36.6	-7.3	37.1		
(5)			-5.9	27.0	1.1	0.7	-6.7	27.5	-7.3	27.9	-8.0	28.2	-8.8	28.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.6	5.9	6.8	9.9	14.6	19.1	22.5	26.5	27.7	24.4	19.0	13.4	8.5
	DBStd	7.88	2.27	2.75	3.05	3.09	2.25	2.23	2.31	1.91	2.70	2.81	2.88	2.80
	HDD10.0	335	128	96	40	3	0	0	0	0	0	0	4	64
	HDD18.3	1583	384	322	261	116	18	1	0	0	1	26	150	306
	CDD10.0	2740	2	7	38	142	283	374	513	550	431	278	105	17
	CDD18.3	945	0	0	0	4	42	124	254	292	182	45	1	0
	CDH23.3	7832	0	0	1	12	160	599	2366	3118	1380	195	3	0
	CDH26.7	2537	0	0	0	1	9	97	791	1214	400	25	0	0
Wind	WSAvg	2.2	2.0	2.1	2.4	2.5	2.4	2.2	2.4	2.3	2.1	1.9	1.8	2.0
Precipitation	PrecAvg	1892	75	91	147	154	169	216	215	183	260	196	120	67
	PrecMax	2698	168	193	283	311	328	347	472	514	524	754	279	164
	PrecMin	1259	1	17	51	65	71	105	22	17	74	21	9	3
	PrecStd	354	45	47	55	63	62	66	116	124	129	143	65	41
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.2	18.8	21.3	24.8	27.9	30.7	34.0	34.6	32.9	29.1	23.3	19.5
		MCWB	11.5	13.2	13.5	16.5	19.5	23.5	25.5	26.0	24.8	21.5	17.3	15.1
	2%	DB	14.1	16.1	19.2	23.0	26.2	29.0	32.7	33.4	31.4	27.0	21.8	17.3
		MCWB	9.1	11.1	13.2	15.8	18.4	22.3	25.3	25.7	24.1	20.5	16.0	12.2
	5%	DB	12.6	14.1	17.6	21.7	25.1	27.7	31.7	32.5	30.3	25.4	20.4	15.6
		MCWB	7.7	9.1	12.0	15.4	18.2	21.4	25.0	25.4	23.7	19.7	15.2	10.6
	10%	DB	11.3	12.5	16.0	20.5	24.0	26.4	30.6	31.6	29.1	24.1	19.0	14.0
		MCWB	6.4	7.5	10.9	14.6	17.7	21.0	24.6	25.1	23.1	18.7	14.3	9.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.0	14.7	16.2	20.1	22.2	24.9	26.9	27.3	26.1	23.5	19.2	16.5
		MCDB	14.8	17.5	18.5	22.2	24.8	29.1	31.8	32.3	30.0	26.3	21.3	18.7
	2%	WB	10.3	12.0	14.5	18.0	20.7	23.8	26.2	26.7	25.2	22.1	17.6	13.2
		MCDB	13.0	15.2	17.6	20.7	23.7	27.2	30.8	31.5	29.1	25.6	20.2	15.7
	5%	WB	8.3	9.9	13.3	16.7	19.7	23.0	25.7	26.2	24.7	20.8	16.3	11.5
		MCDB	11.6	13.2	16.5	20.2	23.1	26.1	30.0	30.7	28.5	24.4	19.0	14.4
	10%	WB	7.0	8.2	11.8	15.7	18.8	22.1	25.2	25.7	24.1	19.6	15.1	10.0
		MCDB	10.5	11.6	15.1	19.2	22.5	25.4	29.3	30.0	27.9	22.9	18.0	13.3
Mean Daily Temperature Range	5% DB	MDBR	10.0	9.8	9.8	9.5	8.7	7.1	6.7	7.3	7.8	8.6	9.5	10.0
		MCDBR	11.3	11.5	11.4	10.9	10.1	8.6	8.0	8.3	8.8	9.7	10.7	11.1
		MCWBR	7.2	7.5	6.9	5.7	4.8	3.6	3.0	3.1	3.6	4.7	6.2	7.0
	5% WB	MCDBR	10.2	9.8	9.6	8.5	8.0	6.6	7.2	7.5	7.7	8.3	8.9	9.6
MCWBR		7.3	7.4	7.4	5.4	4.8	3.2	2.9	3.0	3.5	4.7	6.1	6.9	
Clear-Sky Solar Irradiance	taub	0.387	0.441	0.506	0.562	0.584	0.611	0.598	0.590	0.529	0.500	0.431	0.383	
	taud	2.213	2.064	1.911	1.828	1.835	1.837	1.918	1.926	2.052	2.051	2.159	2.244	
	Ebn at Noon	790	783	763	743	733	711	718	714	737	715	732	767	
	Edh at Noon	111	144	183	208	209	209	191	186	157	145	117	102	
All-Sky Solar Radiation	RadAvg	2.78	3.34	4.16	4.80	5.17	4.52	4.98	5.04	3.98	3.15	2.61	2.45	
	RadStd	0.18	0.30	0.27	0.50	0.54	0.42	0.75	0.42	0.35	0.33	0.23	0.19	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	+1.13	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (45 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page